

GENERAL SCIENCE — GRADE 10 | NUTRITION IN ANIMALS

Quick Check

Lesson 8: DQB Synthesis & Assessment — Driving Question Resolution and Unit Wrap-Up

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.3: Nutrition in Animals | Lesson 8

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A learner is doing Benedict's test on a sample. Which set of steps is in the correct order?

- A** Put about 2 cm³ of test solution in a test tube, add an equal amount of Benedict's solution and mix, place the tube in a hot water bath for a few minutes, then observe and record colour changes
- B** Add Benedict's solution to a dry test tube, add a few drops of test solution, mix, and record the colour immediately without heating
- C** Put about 2 cm³ of test solution in a test tube, place it in a hot water bath, then add Benedict's solution and record the colour once the tube is cold
- D** Heat the Benedict's solution alone, add about 2 cm³ of test solution, then record the colour change without mixing

After heating a glucose solution with Benedict's solution in a hot water bath, a brick-red precipitate forms. What does this result show?

- A** The solution contains a reducing sugar such as glucose
- B** The solution has no sugar, because the blue colour has been lost
- C** The solution contains protein, because Benedict's solution reacts with all nutrients
- D** The solution contains starch, because starch turns Benedict's solution red

In the practical, a dilute glucose solution was used to represent 'digested ugali'. Why is glucose a suitable choice to represent digested ugali?

- A** Ugali starch is already a reducing sugar, so it gives a positive Benedict's test before any digestion
- B** Ugali contains mainly protein, and glucose is the end-product of protein digestion
- C** Hydrochloric acid in the stomach turns starch into glucose within seconds
- D** Amylase enzymes break the large starch molecules in ugali down into simple sugars such as glucose

Nyama choma is rich in protein. Which statement correctly matches the enzyme pepsin to its region of action and substrate?

- A** Pepsin acts in the mouth and breaks down starch into maltose
- B** Pepsin acts in the small intestine and breaks down fats into fatty acids and glycerol
- C** Pepsin acts in the stomach and breaks down starch into glucose
- D** Pepsin acts in the stomach and breaks down proteins into smaller peptides

In the Explain phase, pairs wrote about what happens when an adaptation of the digestive system fails. Infection with the bacterium *H. pylori* can damage the mucus lining of the stomach. Why can this lead to an ulcer?

- A** The stomach starts to digest starch, which burns a hole in the wall
- B** The villi in the stomach are destroyed, so nutrients cannot be absorbed
- C** The stomach can no longer make any acid, so food stays undigested and forms a sore
- D** The wall of the stomach loses its protection and is damaged by the acid and pepsin in the stomach juice

Sukuma wiki, ugali and nyama choma are broken down into small soluble molecules. How do the villi and microvilli of the ileum help the body absorb these nutrients efficiently?

- A** They make the intestine wall thick so that nutrients cannot escape back into the food
- B** They grind the food into smaller pieces so it can pass through the intestine wall
- C** They increase the surface area of the intestine wall, so more nutrients can be absorbed into the rich network of capillaries
- D** They release hydrochloric acid to dissolve the nutrients so they can enter the blood

A learner's Lesson 1 sentence said, 'Enzymes break down food.' Which upgraded sentence is the most scientifically complete?

- A** Enzymes in the mouth and stomach help to break down food into smaller parts.
- B** Pepsin in the mouth breaks down the proteins in ugali.
- C** The mouth breaks down ugali when we chew it with our teeth.
- D** Salivary amylase in the mouth begins to break down the starch in ugali into maltose.

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).